Mobius & Multiplicative

Multiplicative (p prime, $k \ge 1$):

- $I(p^k) = 1$, $\mathrm{Id}(p^k) = p^k$, $\mathrm{Id}_a(p^k) = p^{ak}$.
- $\chi(p^k) = \left[p^k = 1\right]$.
- $\sigma_a(p^k) = \sum_{i=0}^{k} p^{ai}$.
- $\mu(p^k) = [k = 0] - [k = 1]$.
- $\varphi(p^k) = p^k - p^{k-1}$.

$[P]$: boolean (1 true, 0 false).

Mobius IE: square-free

Square-free $\le n$ ($n \le 10^{12}$):

$$\text{ans} = n - \sum_{i=1}^{\lfloor \sqrt{n} \rfloor} \mu(i) \left\lfloor \frac{n}{i^2} \right\rfloor .$$

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long long ans=n;
for(long long i=1;i*i<=n;i++)
  ans-=mo[i]*(n/(i*i));
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Series (ICPC)

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=0}^{n} c^i = \frac{c^{n+1}-1}{c-1} \;(c \ne 1),$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{i=0}^{\infty} c^i = \frac{1}{1-c} \;(|c| < 1),$$

$$\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}, \qquad \sum_{i=0}^{n} i c^i = \frac{nc^{n+2} - (n+1)c^{n+1} + c}{(c-1)^2},$$

$$H_n = \sum_{i=1}^{n} \frac{1}{i}, \qquad \sum_{i=1}^{n} i H_i = \frac{n(n+1)}{2} H_n - \frac{n(n-1)}{4}$$

$$\sum_{i=1}^{n} H_i = (n+1)H_n - n.$$

Combinatoria (ICPC)

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}, \qquad \binom{n}{k} = \binom{n}{n-k}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}, \qquad \binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}.$$

$$\binom{n}{m}\binom{m}{k} = \binom{n}{k}\binom{n-k}{m-k}.$$

$$\sum_{k=0}^{n} \binom{r+k}{k} = \binom{r+n+1}{n},$$

$$\sum_{k=0}^{n} \binom{k}{m} = \binom{n+1}{m+1},$$

$$\sum_{k=0}^{n} \binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}.$$

$$\left\{ {n \atop k} \right\} = k \left\{ {n-1 \atop k} \right\} + \left\{ {n-1 \atop k-1} \right\}, \qquad C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Burnside & Pőlya

Cuenta clases bajo simetrías. G actúa, $I(\pi)$=fijos, $C(\pi)$=#ciclos.

$$|\text{orb}| = \frac{1}{|G|} \sum_{\pi \in G} I(\pi),$$

$$|\text{orb}| = \frac{1}{|G|} \sum_{\pi \in G} k^{C(\pi)}.$$

CSES 2209: Necklaces

Rotaciones en n perlas, k colores, $G = C_n$:

$$\boxed{\frac{1}{n} \sum_{i=0}^{n-1} k^{\gcd(i,n)} = \frac{1}{n} \sum_{d \mid n} \varphi\!\left(\frac{n}{d}\right) k^d}.$$

CSES 2210: Grids

Binaria $n \times n$, rotaciones C_4, $k = 2$.

$$a = \left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n+1}{2} \right\rfloor, \qquad o = n \bmod 2.$$

$$\boxed{\frac{1}{4}\left(2^{n^2} + 2^{2a+o} + 2 \cdot 2^{a+o}\right)}.$$

CF 1065E: Transmutation

Longitud n, alfabeto k. Ops por b_i:

$$1 \le b_1 < \cdots < b_m, \; d_1 = b_1, \; d_i = b_i - b_{i-1}.$$

Solo extremos $2b_m$, centro libre k^{n-2b_m}. Cada capa d_i: id k^{2d_i}, swap k^{d_i}. Grupo $|G| = 2^m$. Burnside:

$$\boxed{k^{n-2b_m} \prod_{i=1}^{m} \frac{k^{2d_i} + k^{d_i}}{2}}$$

$$\boxed{k^{n-b_m} \cdot \frac{1}{2^m} \prod_{i=1}^{m} (1 + k^{d_i})}.$$

Geometry

Frustum (glass)

Water height p, radii r_1, r_2:

$$V = p\pi \frac{r_1^2 + r_2^2 + r_1 r_2}{3}.$$

Sine/Cosine

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \quad c^2 = a^2 + b^2 - 2ab\cos C.$$

Theorems

Erdős–Szekeres

Si $n = ab + 1$, toda secuencia longitud n tiene inc. de $a + 1$ o dec. de $b + 1$. (Dilworth generaliza.)

Grundy

$g(\text{terminal}) = 0$, $g(v) = \text{mex}\{g(u)\}$, juegos indep.: $g = g_1 \oplus \cdots \oplus g_n$.

Főrmulas clave

Cuadrãtica: $ax^2 + bx + c = 0 \; (a \ne 0), \; x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Binomial: $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$.

Lucas (p primo): $\binom{n}{k} \equiv \prod_{i \ge 0} \binom{n_i}{k_i} \pmod{p}$,

$$n = \sum n_i p^i, \; k = \sum k_i p^i, \; 0 \le n_i, k_i < p.$$